
EE 614: Optical Communication Systems

Course Code : EE 614

Course Name : Optical Communication Systems

L-T-P-C : 3-0-0-3

Prerequisites : EE-304: Communication Theory and the instructors consent

Intended for : UG/MS/PhD

Core or Elective : Elective

Approval: 5th Senate

Course Contents

- **INTRODUCTION TO OPTICAL COMMUNICATION AND FIBER CHARACTERISTICS:**
 - Evolution of Light wave systems, System components, Optical fibers, Step Index & Graded index Mode theory, Fiber modes, Dispersion in fibers, Limitations due to dispersion, Dispersion shifted and dispersion flattened fibers, Fiber Losses and Non-linear effects. [8 Lectures]
- **OPTICAL TRANSMITTERS:**
 - Basic concepts, LED's structures, Spectral Distribution, Semiconductor lasers, Structures, Threshold Conditions, and Transmitter design. [4 Lectures]
- **OPTICAL DETECTORS AND AMPLIFIERS:**
 - Basic Concepts, PIN and APD diodes structures, Photo detector Noise, Signal impairments, Receiver design. Amplifiers: Basic concepts, Semiconductor optical amplifiers, Raman, Brillouin amplifiers, Erbium doped fiber amplifiers, pumping requirements, cascaded in-line amplifiers. [8 Lectures]
- **COHERENT LIGHTWAVE SYSTEMS:**
 - Basic coherent systems, Coherent detection principles, Homodyne and heterodyne detection, Modulation formats, BER in synchronous receivers, Equalization, carrier phase and frequency synchronization, timing synchronization. [4 Lectures]
- **MULTICHANNEL SYSTEMS:**
 - WDM Light wave Systems, WDM Components, WDM System Performance Issues, Time- Division Multiplexing, Subcarrier Multiplexing, Orthogonal Frequency Division Multiplexing (OFDM) and Code-Division Multiplexing. [6 Lectures]
- **OPTICAL TRANSMISSION LINK LIMITS:**
 - Power budget and bandwidth limited point-to-point light wave system, OSNR evaluation in high speed optical transmission systems, Dispersion Management, Nonlinearity management. [6 Lectures]
- **OPTICAL NETWORKS:**
 - LANs, MANs, Long-Haul Networks, Design Guidelines. [6 Lectures]



Reference Books:

1. John M. Senior, **Optical Fiber Communications: Principles and Practice**, 2nd Edition, Prentice Hall of India.
2. G. P. Agrawal, **Fiber Optic Communication Systems**, 3rd Edition, John Wiley & Sons, 2002.
3. G. Keiser, **Optical Fiber Communication Systems**, McGraw Hill, 2000.
4. M. Cvijetic and Ivan Djordjevic, **Advanced Optical Communication Systems and Networks**, Artech House, 2013.
5. Ramaswami, Sivarajan, and Sasaki, **Optical networks: A practical perspective**, 3rd Edition, Morgan-Kauffman, 2009.